

Ex 1)

The CoM of the first link is $p_1 = \begin{cases} d_1 c_1 \\ d_1 s_1 \end{cases}$ so $\dot{p}_1 = \begin{cases} -d_1 s_1 \dot{q}_1 \\ d_1 c_1 \dot{q}_1 \end{cases}$

$$\Rightarrow T_1 = \frac{1}{2} \dot{q}_1^2 (I_1 + m_1 d_1^2) \quad \text{and} \quad p_2 = \begin{cases} l_1 c_1 + q_2 \cos(q_1 - \frac{\pi}{2}) \\ l_1 s_1 + q_2 \sin(q_1 - \frac{\pi}{2}) \end{cases} = \begin{cases} l_1 c_1 + q_2 s_1 \\ l_1 s_1 - q_2 c_1 \end{cases}$$

$$\Rightarrow \dot{p}_2 = \begin{cases} -l_1 s_1 \dot{q}_1 + q_2 c_1 \dot{q}_1 + \dot{q}_2 s_1 \\ l_1 c_1 \dot{q}_1 + q_2 s_1 \dot{q}_1 - \dot{q}_2 c_1 \end{cases} \quad T_2 = \frac{1}{2} I_2 \dot{q}_1^2 + \frac{1}{2} m_2 [(l_1^2 + q_2^2) \dot{q}_1^2 + \dot{q}_2^2 - 2 l_1 \dot{q}_1 \dot{q}_2]$$

$$M_{11} = I_1 + m_1 d_1^2 + I_2 + m_2 l_1^2 + m_2 q_2^2 \quad M_{12} = -m_2 l_1 \quad M_{22} = m_2$$

$\begin{matrix} e_1 + e_2 q_2^2 \\ -e_3 \\ e_2 \end{matrix}$

$$M = \begin{bmatrix} e_1 + e_2 q_2^2 & -e_3 \\ -e_3 & e_2 \end{bmatrix} \quad \text{Now let's derive } c(q, \dot{q})$$

$$C_1 = \frac{1}{2} \left(\frac{\partial M_{11}}{\partial q} + \left(\frac{\partial M_{11}}{\partial q} \right) - \frac{\partial M}{\partial q_1} \right) = e_2 q_2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Rightarrow C_1 = \dot{q}^T C_1 \dot{q} = 2 e_2 q_2 \dot{q}_1 \dot{q}_2$$

$$C_2 = \frac{1}{2} \left(\frac{\partial M_{22}}{\partial q} + \left(\frac{\partial M_{22}}{\partial q} \right) - \frac{\partial M}{\partial q_2} \right) = - \begin{pmatrix} e_2 q_2 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow C_2 = -e_2 q_2 \dot{q}_1^2$$

Now, find $g(q)$. $U = g_0 (d_1 s_1 m_1 + m_2 [l_1 s_1 - q_2 c_1]) =$
 $= g_0 [s_1 (d_1 m_1 + l_1 m_2) - q_2 c_1 m_2] = g_0 [s_1 e_4 - q_2 c_1 e_2]$

$$g = \nabla_q U = \begin{cases} g_0 [c_1 e_4 + q_2 s_1 e_2] \\ -g_0 c_1 e_2 \end{cases} \quad \text{The model is:}$$

$$\begin{cases} (e_1 + e_2 q_2^2) \ddot{q}_1 - e_3 \ddot{q}_2 + 2 e_2 q_2 \dot{q}_1 \dot{q}_2 + g_0 [c_1 e_4 + q_2 s_1 e_2] = \tau_1 \\ -e_3 \ddot{q}_1 + e_2 \ddot{q}_2 - e_2 q_2 \dot{q}_1^2 - g_0 c_1 e_2 = \tau_2 \end{cases}$$

$$\begin{bmatrix} \ddot{q}_1 & \ddot{q}_1 q_2^2 + 2 q_2 \dot{q}_1 \dot{q}_2 + g_0 q_2 s_1 & -\ddot{q}_2 & g_0 c_1 \\ 0 & \ddot{q}_2 - q_2 \dot{q}_1^2 - g_0 c_1 & -\ddot{q}_1 & 0 \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \end{bmatrix} = \tau$$

$\nearrow Y$ is the regressor matrix, It's column are lin. ind. hence 4 is the minimal number of dynamic coefficients.

Ex 2) The generalized momentum is :

$$p = M\dot{q} = \begin{cases} (e_1 + e_2 K^2) \dot{q}_1 - e_3 \dot{q}_2 \\ -e_3 \dot{q}_1 + e_2 \dot{q}_2 \end{cases}, \quad \text{and} \quad \dot{q}(t) = \begin{cases} \frac{\pi}{T} \sin\left(\frac{\pi}{T}t\right) \\ 0 \end{cases} \quad \text{hence}$$

$$* \int \sin(\omega t) = -\frac{1}{\omega} \cos(\omega t)$$

$$p(t) = \begin{cases} (e_1 + e_2 K^2) \frac{\pi}{T} \sin\left(\frac{\pi}{T}t\right) \\ -e_3 \frac{\pi}{T} \sin\left(\frac{\pi}{T}t\right) \end{cases} \quad \text{we have to integrate both quantities:}$$

$$h_1 = \int_0^T (e_1 + e_2 K^2) \frac{\pi}{T} \sin\left(\frac{\pi}{T}t\right) dt = (e_1 + e_2 K^2) \frac{\pi}{T} \int_0^T \sin\left(\frac{\pi}{T}t\right) dt =$$

$$(e_1 + e_2 K^2) \frac{\pi}{T} \left[-\frac{T}{\pi} \cos(\pi) + \frac{T}{\pi} \cos(0) \right] = 2(e_1 + e_2 K^2)$$

$$h_2 = \int_0^T -e_3 \frac{\pi}{T} \sin\left(\frac{\pi}{T}t\right) dt = -e_3 \frac{\pi}{T} \left(2 \frac{T}{\pi} \right) = -2e_3$$

Ex 3) The objective function is: $(M\dot{q})^T \dot{q} = \dot{q}^T M \dot{q}$, so we want to minimize the kinetic energy. This, can be obtained by considering $J_M^\#$ (weighted pseudo inverse), but we want to address $p_2 = c$. First of all notice how

$$p_2 = M_2 \dot{q} \quad \text{where} \quad M_2 \text{ are the first 2 rows of } M.$$

So $M_2 \dot{q} = c$, we use the weighted pseudo inverse of

$$M_2 \Rightarrow \dot{q} = (M_2)^\#_M c = \tilde{M}^{-1} M_2^T (M_2 \tilde{M}^{-1} M_2^T)^{-1} c.$$

For the RP robot we have

$$(M_2)^\#_M c = \frac{c}{e_2 q_2^2 + e_1} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Ex 4) This is an impedance control problem, we are requiring the model to have a desired impedance dynamics. First we must linearize the system, let

$$Z = J^T (M_Z \ddot{z} + C_Z \dot{z} + g_Z - F_Z)$$

where $F_Z = (F_d \ 0)^T$ is the measured Force, and M_Z, C_Z, g_Z the model in task space. This achieves the following: $\ddot{p} = \ddot{z}$ with $p = (x \ z)^T = K(q)$.

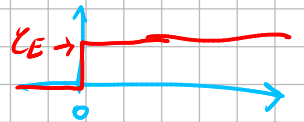
$$\begin{cases} \ddot{z} = \ddot{z}_x \\ \ddot{z} = \ddot{z}_z \end{cases} \quad \text{so we let } z_z = -K_D \dot{z} - K_P (z - z_E), \text{ in this way}$$

$$\ddot{z} + K_D \dot{z} + K_P (z - z_E) = 0. \text{ Then we complete } z_x$$

$$z_x = \frac{1}{M_d} (F_d - D_d \dot{z}_x - K_d (x - x_E)) \Rightarrow$$

$M_d \ddot{z}_x + D_d \dot{z}_x + K_d (x - x_E) = F_d$. Now we determine the final eq. position for both the decoupled axes.

$$\ddot{z} + K_D \dot{z} + K_P (z - z_E) = 0 \Rightarrow \text{Step input at } t=0$$



$$z s^2 + K_D z s + K_P z = K_P z_E \frac{1}{s} \Rightarrow$$

$$z (s^2 + K_D s + K_P) = z_E K_P \frac{1}{s} \Rightarrow z = \frac{z_E K_P \frac{1}{s}}{(s^2 + K_D s + K_P)} \quad \lim_{s \rightarrow 0} s z(s) = \frac{K_P}{K_P} z_E = z_E$$

So the final position of z will be z_E . Let's see x

$$M_d \ddot{x} + D_d \dot{x} + K (x - x_E) = F_d \Rightarrow$$

$$M_d x s^2 + D_d x s + K x = K x_E \frac{1}{s} + F_d \Rightarrow x = \frac{K x_E \frac{1}{s} + \frac{F_d}{s}}{M_d s^2 + D_d s + K}$$

$$\lim_{s \rightarrow 0} s x(s) = x_E + \frac{F_d}{K_d}$$

Ex 5)

$$\begin{cases} (e_1 + e_2 q_2^2) \ddot{q}_1 - e_3 \ddot{q}_2 + 2e_2 q_2 \dot{q}_1 \dot{q}_2 + g_0 [c_1 e_1 + q_2 s_1 e_2] = \tau_1 \\ -e_3 \ddot{q}_1 + e_2 \ddot{q}_2 - e_2 q_2 \dot{q}_1^2 - g_0 c_1 e_2 = \tau_2 \end{cases}$$

$$\begin{cases} \tau_1 = (e_1 + e_2 K_2^2) b \frac{\pi^2}{T^2} \cos\left(\frac{\pi}{T} t\right) \\ \tau_2 = -e_3 b \frac{\pi^2}{T^2} \cos\left(\frac{\pi}{T} t\right) - e_2 K \left[b \frac{\pi}{T} \sin\left(\frac{\pi}{T} t\right) \right]^2 \end{cases}$$

$\tau_1(t)$ is maximum in $t=0 \Rightarrow \max \tau = (e_1 + e_2 K_2^2) b \frac{\pi^2}{T^2}$, this should be not more than $\tau_{max,1}$ so

$$(e_1 + e_2 K_2^2) b \frac{\pi^2}{T^2} = \tau_{max,1} \Rightarrow \frac{1}{T^2} = \frac{\tau_{max,1}}{(e_1 + e_2 K_2^2) b \pi^2} \Rightarrow T^* = \sqrt{\frac{(e_1 + e_2 K_2^2) b \pi^2}{\tau_{max,1}}}$$

in $t=0$ then $\tau_2(t) = \frac{-e_3 \tau_{max,1}}{(e_1 + e_2 K_2^2)}$